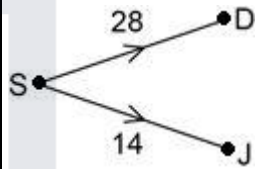
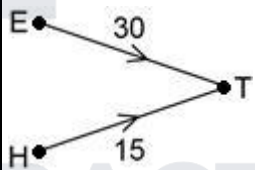


Mark schemes

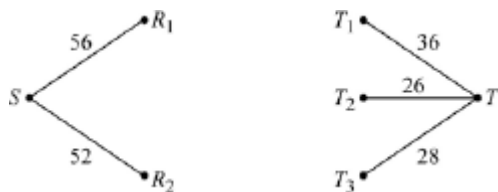
Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds the value of the cut	AO1.1b	B1	37
(i)				
(ii)	Deduces a weak inequality for the value of max flow based on their value of the cut Must see $\leq$ their value of the cut OE	AO2.2a	B1F	Max flow ≤ 37
(b)	Lists both sources CAO	AO1.1b	B1	D and J
(i)				
(ii)	Adds supersource S with directed arcs to D and J labelled with weights at least 28 and at least 14 respectively	AO1.1b	B1	
(c)	Lists both sinks CAO	AO1.1b	B1	E and H
(i)				
(ii)	Adds supersink T with directed arcs to E and H labelled with weights at least 30 and at least 15 respectively	AO1.1b	B1	
				Total 6 marks

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Q2.

(a)



Edges with values $\geq 56, 52$

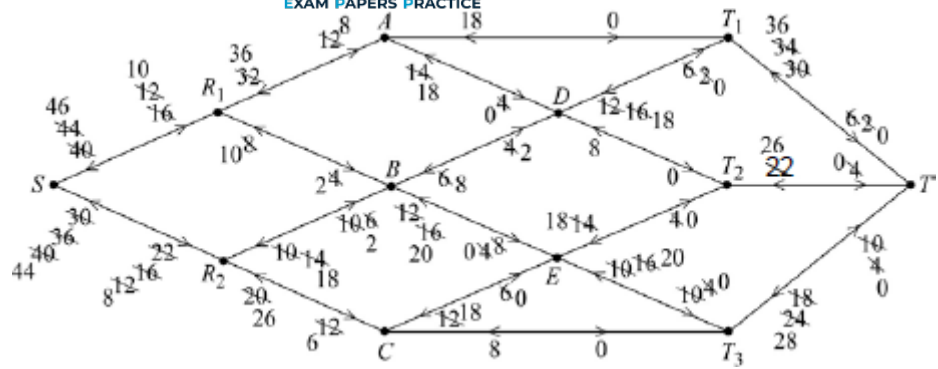
B1

Edges with values $\geq 36, 26, 28$

B1

2

(b)(i)



initial diagram with forward / back flows

M1

Fully correct diagram

A1

$S R_1 A D T_1 T$ 4

One correct path and flow

M1

$S R_1 B D T_1 T$ 2

At least one other correct path and flow

A1

$S R_2 C E T_3 T$ 6

$S R_2 B E T_2 T$ 4

all correct

A1

$S R_2 B E T_3 T$ 4

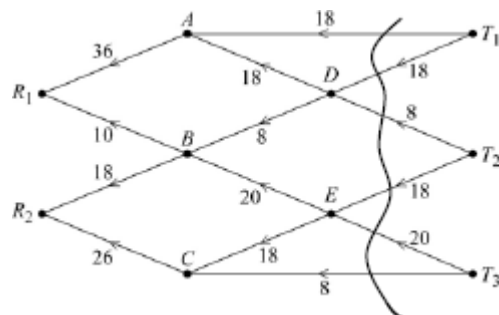
(ignore connections to S and T)

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5

(ii) Max flow 90

B1



B1

2

- (c) Cut through (shown)
PI by correct list

B1

$AT_1, DT_1, DT_2, ET_2, ET_3, CT_3$

B1

OE

2

[11]

Q3.

- (a) $SP \geq 12$
 $SQ \geq 10$
 $SR \geq 17$

S in correct place, (arrows) and capacities

B1

$YT \geq 18$
 $ZT \geq 17$

T in correct place, (arrows) and capacities

B1

2

- (b) $SPUYT$ 10

B1

$SRVWZT$ 8

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2

- (c) (i) Initial flow forward and backward

PU 2 and 10 ; UY 0 and 10

B1

RV 0 and 8 ; VW 1 and 8 ; WZ 2 and 8

withhold one B1 if paths to S and T not updated

B1

2

- (ii) Two correct routes and flows on Figure 2

<i>SPUYT</i>	10
<i>SRYWZT</i>	8
<i>SPUXYT</i>	2
<i>SQVUXYT</i>	6
<i>SRWXZT</i>	5
<i>SRWZT</i>	2

(other possibilities)

M1

Correct additional flows
Max flow = 33

A1

Adjustment of at least 4 edges corresponding to flows
(forward and backward)

M1

Correct final flows forward and backward
(must score A1 for table)

edges UY, UX, WX and WZ will be saturated

$XY + XZ = 13$ in back flow

A1cso

4

- (d) Cut with value 33 is through
UY, UX, WX and WZ

B1

1

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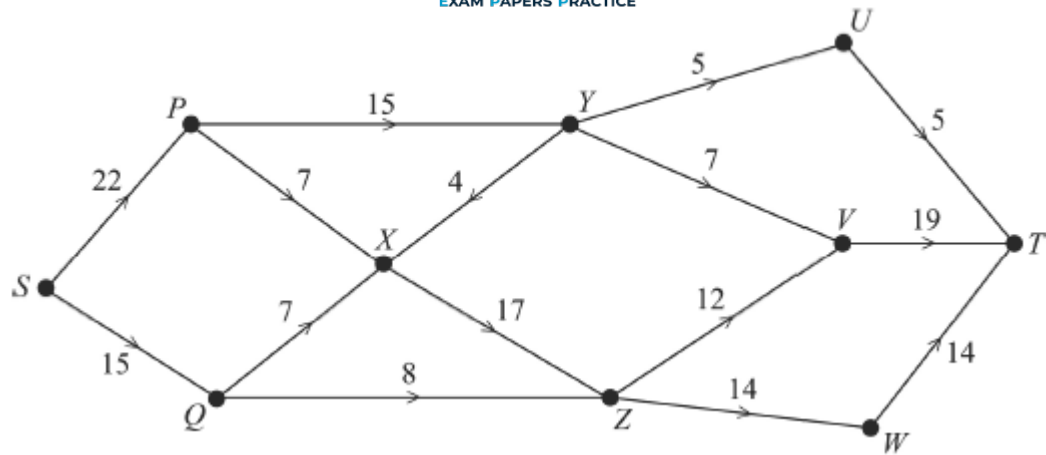
Q4.

- (a) Correct position of *S* and *T*

M1

Values on edges *SP, SQ, UT, VT* and *WT*

A1



2

- (b) (i) Cut C has value 40
 $15 + 0 + 17 + 8$

B1

1

- (ii) Max flow ≤ 40

E1

1

- (c)

Route	Flow
$SQZWT$	8
$SPYXZVT$	4

B1B1

2

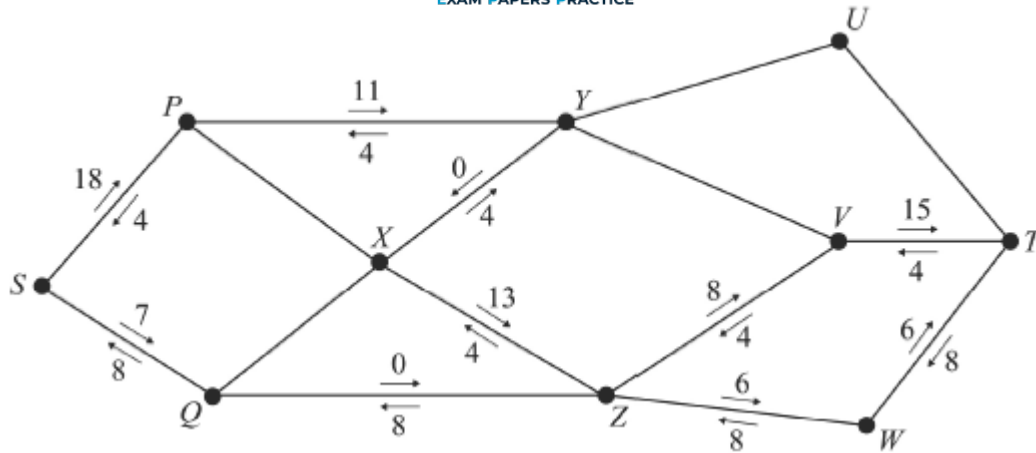
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- (d) (i) 3 forward and backward flows correct

M1

All initial values correct on edges below

A1



2

(ii)

Route	Flow
SQZWT	8
SPYXZVT	4
SPYUT	5
SPYVT	6
SPXZVT	7
SQXZWT	6
SQXYVT	1

(Many different possibilities)
2 or more correct flows in table

M1

Table correct (adding to 37)

A1

At least 2 flows augmented on diagram

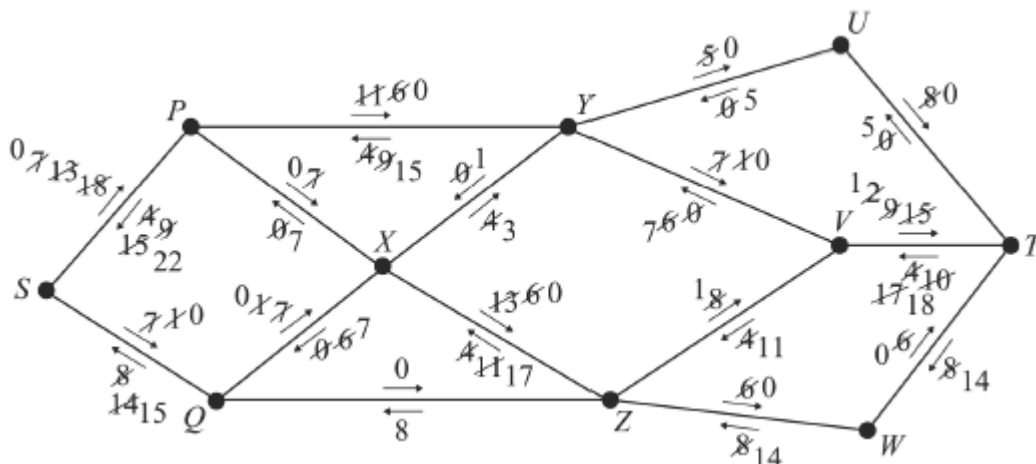
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M1

Correct forward and backward final flows

A1

4



Other possibility for ZV, VT, ZW and WT

- (e) Flow from Y to X is 3

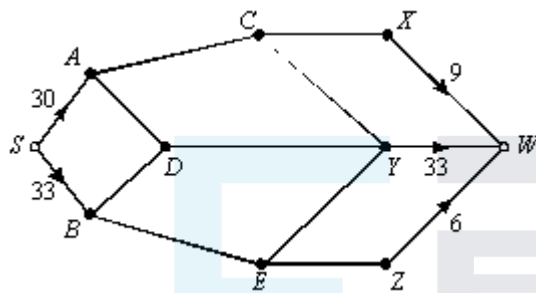
B1

1

[13]

Q5.

(a)



for S and W

for five arcs correct ($\geq$)

2

M1

A1

- (b) $SADYW = 10$

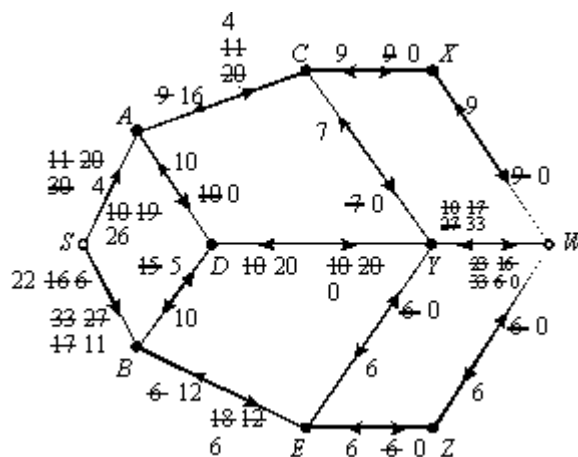
B1

$SBEZW = 6$ Exam Papers Practice. All rights reserved.

B1

2

- (c) (i)



for starting with SADYW, SBEZW

SCA

M1

for AC

M1

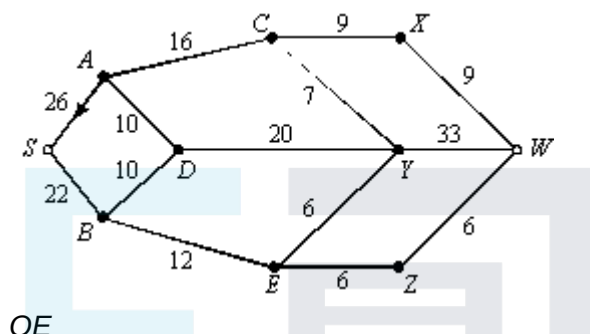
for SA

A1

for SB

A1

A1



(ii) Maximum flow = 48

B1

6

M1

Minimum cut CX, YW, EZ

(9) (33) (6)

= 48

A1

2

[12]

Q6.

(a) Vertices S and T

M1

Arcs SS_1 , SS_2 , T_1T , T_2T and T_3T

Capacities 18, 15, 10, 13, 12 (or more) respectively

A1

2

(b) (i) $8 + 2 + 5 + 4 + 12 = 31$

B1

1

(ii) $AB \ AC \ DC \ DE$

M1

(or $AB \ CB \ CT_2 \ CE \ DE$)

A1

2

(c) e.g. $SS_1ABT_1T: 8$

M1A1

$SS_1DET_3T: 7$

A1

$SS_2DCET_3T: 5$

A1

$SS_2DACT_2T: 5$

A1

$SS_1ACBT_2T: 2$

$SS_1DCET_2T: 3$

A1

6

(d) All flows $\leq$ all cuts

M1

So, by (b)(ii), all flows ≤ 30 .

Hence the flow of 30 is maximum possible.

A1

2

(e) e.g. For T_1 to get 10 BT_1 will have a flow of 10.

M1

Then, looking at vertex B , max inflow = 10.

Hence BT_2 has 0 flow.

A1

So maximum arriving at T_2 is from CT_2
and ET_2 with a total capacity of 9.

A1

3

[16]

Q7.

- (a) Vertex S , arcs SS_1 and SS_2 with

M1

capacities at least 10 and 11 respectively.

A1

2

- (b) e.g. SS_1ABT 4

M1

SS_1CDT 5

A1

SS_2CDBT 3

A1

SS_2EFT 6

A1

SS_2CEFT 2

A1

5

- (c) S_1A, CD, EF

M1

Is a cut of 20.

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A1

So all flows ≤ 20

A1

3

[10]

Examiner reports

Q1.

The majority of students determined the value of the cut correctly, with the most common errors being numerical slips. Only 51% of students correctly interpreted their value of the cut, with many stating generic comments about the maximum-flow minimum-cut theorem but not making the connection between the value of any cut and the implications for the maximum flow.

Most students spotted the sources and sinks of the network, the most common error being to include a third node which was not a source or sink. However, less than half of the students correctly added arcs to the diagram, with common mistakes being the lack of arrows or correct capacities on each arc.

Q2.

Although many students answered this part correctly, there was a significant number who included values on their edges that corresponded to the initial flow. In part (b), a large number of students started from a 'zero-flow' and then augmented to arrive at the given flow and then stopped. In (b)(i), although there were many correct solutions, a significant number of students failed to give a convincing solution on their diagram. It is essential that students use 'backward' arrows to show existing flows and 'forward' arrows to show potential flows. There were a significant number of students who scored 5 marks in part (b)(i) but failed to score both marks part (ii). Part (c) was poorly answered. Students were asked to list the edges of their cut, many simply drew a cut on their diagram.

Q3.

In part (a) most candidates understood how to add a super source and super sink to the network and inserted correct weights on the edges.

In (b) again, few candidates had trouble in finding the maximum flows along the given edges. It was not clear whether some had misread the second route when they gave the flow as 9 instead of 8. For this reason, this error was not penalised heavily in the subsequent work if they eventually found the correct maximum flow through the network.

In part (c) it was good to see candidates trying to set out their solution in a logical manner and once again the diagram and table clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. Marks are awarded for the initial flow and it is very difficult to credit candidates for their original values if they have been obliterated during augmentation.

In (d) those who had a correct maximum flow in part (c) were usually able to find the correct cut with value equal to the maximum flow.

Q4.

Part (a) All candidates seemed familiar with the idea of a super-source, S , and super-sink, T . Although it was possible to assign equal values such as 100 to the capacity on each of the new edges, such a large value often caused problems with flow augmentation. Some incorrectly assigned equal values such as 20 to the new edges. Those using capacities of

22 on *SP* and 15 on *SQ* were usually the most successful.

Part (b) Quite a few candidates made errors in calculating the value of the cut with a common wrong answer being 36. Several incorrectly thought that the maximum flow was equal to the value of this cut.

Part (c) The values of the initial maximum flows along the given edges were usually correct.

Part (d) It was good to see candidates trying to set out their solution in a logical manner with most candidates showing potential forward and backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values, particularly the initial flow, should be clear to the examiner when such modification is made. It is not wise to totally obliterate previous values when adjustments are made.

Part (e) Some candidates misread this last part and stated the maximum flow through the network. Very few were able to use their final maximum flow diagram to interpret the correct flow from Y to X.

Q5.

Candidates were required to add a super source and a super sink to the given diagram. This was well answered although a number of candidates did not include values on the arcs that at least equalled the total values at points *A* and *B*. Part (b) was well answered and to assist candidates they were expected to take these as initial flows in part (c) before using flow augmentation to the diagram. There were many excellent solutions to this question, with working shown on the diagrams, together with a clearly labelled final solution.

Q6.

Although candidates generally understood what was required, the presentation of the answers was very muddled and hard to follow. In this sort of question, candidates should clearly indicate their final answer, possibly by ringing the flow in each arc or by drawing a separate figure for the final answer.

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